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Analysis of genetic diversity and phylogenetic relationships among red jungle fowls and Chinese domestic fowls

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Genetic diversity and phylogenetic relationships among 568 individuals of two red jungle fowl subspecies (*Gallus gallus spadiceus* in China and *Gallus gallus gallus* in Thailand) and 14 Chinese domestic chicken breeds were evaluated with 29 microstaellite loci, the genetic variability within population and genetic differentiation among population were estimated, and then genetic diversity and phylogenetic relationships were analyzed among red jungle fowls and Chinese domestic fowls. A total of 286 alleles were detected in 16 population with 29 microsatellite markers and the average number of the alleles observed in 29 microsatellite loci was 9.86±6.36. The overall expected heterozygosity of all population was 0.6708±0.0251, and the number of population deviated from Hardy-Weinberg equilibrium per locus ranged from 0 to 7. In the whole population, the average of genetic differentiation among population, measured as *FST* value, was 16.7% (*P*<0.001), and all loci contributed significantly (*P*<0.001) to this differentiation. It can also be seen that the deficit of heterozygotes was very high (0.015) (*P*<0.01). Reynolds' distance values varied between 0.036 (Xiaoshan chicken-Luyuan chicken pair) and 0.330 (*G. gallus gallus*-Gushi chicken pair). The *Nm* value ranged from 0.533 (between *G. gallus gallus* and Gushi chicken) to 5.833 (between Xiaoshan chicken and Luyuan chicken). An unrooted consensus tree was constructed using the neighbour-joining method and the Reynolds' genetic distance. The heavy-body sized chicken breeds, Luyuan chicken, Xiaoshan chicken, Beijing Fatty chicken, Henan Game chicken, Huainan Partridge and Langshan chicken formed one branch, and it had a close genetic relationship between Xiaoshan chicken-Luyuan chicken pair and Chahua chicken-Tibetan chicken pair. Chahua chicken and Tibetan chicken had closer genetic relationship with these two subspecies of red jungle fowl than other domestic chicken breeds. *G. gallus spadiceus* showed closer phylogenetic relationship with Chinese domestic chicken breeds than *G. gallus gallus*. All 29 microstaellite loci in this study showed high levels of polymorphism and significant genetic differentiation was observed among two subspecies of red jungle fowl and 14 Chinese domestic chicken breeds. The evolutional dendrogram is as follows: evolutional breeds→primitive breeds (Chahua chicken and Tibetan)→red jungle fowl in China (*G. gallus spadiceus*)→red jungle fowl in Thailand (*G. gallus gallus*). The results supported the theory that the domestic fowls might originate from different subspecies of red jungle fowl and Chinese domestic fowls had independent origin.

red jungle fowl, domestic fowl, microsatellite, genetic diversity, phylogenetic relationship

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Since Darwin suggested that the wild progenitor of the domestic chicken belongs to the genus *Gallus*, this hypothesis has been generally accepted. China has a wide variety of indigenous poultry resources with its long history of animal husbandry and diversified geographical conditions. Analysis of genetic diversity and phylogenetic relationships among red jungle fowls and Chinese domestic fowls could provide a theoretical foundation for the research on the origin, evolution, conservation and reasonable utilization of chicken breeds. Some researchers have studied the phylogenetic relationships among red jungle fowls and Chinese domestic fowls based on the results of morphology, archaeological discoveries, plasma protein polymorphic loci and DNA fingerprints[1−4]. However, it still remains controversial on whether two red jungle fowl subspecies (*Gallus gallus spadiceus* in China and *Gallus gallus gallus* in Thailand) can be identified as the same subspecies, and whether Chinese domestic fowls derived from red jungle fowl in Thailand and its adjacent region or have independent origin[4−6]. With the characteristics of high polymorphism, locus specificity, abundance and random distribution over the genome, and their codominant inheritance, microsatellites are currently most commonly used to assess population structure and diversity[7,8]. But research on genetic diversity of red jungle fowls and their phylogenetic relationships with domestic fowls focused on the analysis of partial fragments of mitochondrial DNA[4], and there have no public reports on genetic diversity and phylogenetic relationships among red jungle fowls and Chinese domestic fowls by using microsatellite markers until now. Genetic diversity and phylogenetic relationships among 568 individuals of two red jungle fowl subspecies (*G. gallus spadiceus* in China and *G. gallus gallus* in Thailand) and 14 Chinese domestic chicken breeds were evaluated with 29 microstaellite loci, the genetic variability within population and genetic and phylogenetic relationships among red jungle fowls and Chinese indigenous chicken breeds were analyzed. The results may be useful to understanding origin of domestic chickens and genetic differentiation of these important local breeds, and contribute to a more efficient conversation strategies.

1 Materials and methods

1.1 Experimental populations

A total of 568 individuals originating from two red jungle fowl subspecies (*G. gallus spadiceus* in China and *G. gallus gallus* in Thailand) and 14 Chinese domestic chicken breeds were genotyped in this study. All populations except for Wannan Three-yellow chickens, Huainan Partridges and red jungle fowls were from the Poultry Institute, Chinese Academy of Agricultural Sciences, Yangzhou, China; Wannan Three-yellow Chickens were from a poultry resource center in Qinyan County, Anhui Province; Huainan Partridges were from a poultry resource center in the Institute of Agricultural Science in Huainan city, Anhui Province; the red jungle fowls (*G. gallus spadiceus*) were from the Wild Animal Conservation Centre, Yunnan Province, China. Twenty-six DNA samples of red jungle fowls (*G. gallus gallus*) were presented by Dr. Steffen Weigend, member of the AVIANDIV group. Information about the 16 populations, abbreviation and number of individuals sampled are presented in Table 1.

1.2 DNA isolation

For each individual, 0.4 mL whole blood was collected from the ulnar vein with heparin as anticoagulant. Then, 4 mL of DNA lysate solution (2 mol/L urea, 100 mmol/L Tris-HCl (pH 8.0), 1% SDS, 100 mmol/L EDTA) was added, and the mixture was stored at 4℃. DNA was isolated by using a phenol/chloroform-based method[9].

1.3 Genotyping

The DNA polymorphism was assessed at 29 microsatellite loci (Table 2). These markers were randomly distributed across the chicken genome and recommended by FAO. Several multiplex PCRs were carried out, with 2−5 pairs of primers involved in each reaction. Each PCR tube contained 20 ng of genomic DNA, 10 pmol of each forward primer labeled with either IRD700 or IRD800 (MWG-Biotech, Ebersberg, Germany), 10 pmol of each unlabeled reverse primer, and 1 mmol/L tetramethylammoniumchloride. The amplification protocol comprised of an initial denaturation and enzyme activation phase at 95℃ (15 min), followed by 35 cycles of denaturation at 95℃ (1 min), primer annealing at temperature varying between 58℃ and 64℃ (1 min), and extension at 72℃ (1 min), and a final extension at 72℃ for 10 min. DNA fragments were scored on 8% polyacrylamide gel using an LI-COR automated DNA analyzer (LI-COR Biotechnology Division, Lincoln, NE). Electrophoregram processing and allele-size scoring was

Table 1 Description of the 16 chicken populations

Breed (abbreviation)	Main original area	Specific features	No. of animals studied
Xianju chicken (XJ)	Xianju county, Zhejiang	three yellow[a)],light-sized, layer breed	38
Chahua chicken (CH)	Xishuangbanna, Yunnan	light-sized, meat and egg dual-purpose breed	38
Luyuan chicken (LY)	Zhangjiagang city, Jiangsu	heavy-sized, meat and egg dual-purpose breed	34
Gushi chicken (GS)	Gushi county, Henan	three yellow[a)], medium-sized, meat and egg dual-purpose breed	40
Tibetan chicken (TC)	Ganzi and Aba Tibetan autonomous region	light-sized, selected for yellow plumage, meat and egg dual-purpose breed	38
Baier chicken (BE)	Shangrao city, Jiangxi	three yellow[a)], light-sized, layer breed, white earlobe	34
Dagu chicken (DG)	Zhuanghe county, Liaoning	heavy-sized, meat and egg dual-purpose breed	34
Henan game (HG)	Zhengzhou city, Henan	heavy-sized, fancy breed	32
Langshan chicken (LS)	Rudong county, Jiangsu	heavy-sized, meat and egg dual-purpose breed	40
Taihe silkies (TS)	Taihe county, Jiangxi	light-sized, medicine and entertainment breed	40
Xiaoshan chicken (XIS)	Xiaoshan county, Zhejiang	heavy-sized, meat and egg dual-purpose breed	40
Beijing Fatty chicken (BF)	Chaoyang, Beijing	heavy-sized, meat and egg dual-purpose breed	38
Huainan Partridge (HP)	Huainan city, Anhui	heavy-sized, meat and egg dual-purpose breed	32
Wannan Three-yellow (WTY)	Qinyan county, Anhui	medium-sized, egg purpose breed	32
Gallus gallus spadiceus (GGS)	Shimao county, Yunnan	red jungle fowl (wild)	32
Gallus gallus gallus (GGG)	Middle and western mountainous area, Thailand	red jungle fowl (wild)	26

a) Three yellow features (plumage yellow, beak yellow and shank yellow).

Table 2 Names, chromosome location, total number of alleles, size of alleles, expected heterozygosity and *PIC* of 29 microstellite markers

Locus	Chromosome	No. of alleles	Size of alleles (bp)	Expected heterozygosity	*PIC*
MCW0103	Chr.3	2	266－270	0.4506	0.29
MCW0216	Chr.13	8	137－149	0.5290	0.27
MCW0295	Chr.4	12	88－110	0.7177	0.57
ADL0278	Chr.8	12	114－129	0.6704	0.44
MCW0222	Chr.3	4	220－226	0.6080	0.46
MCW0037	Chr.3	6	154－159	0.6581	0.43
ADL0268	Chr.1	8	104－118	0.7180	0.53
MCW0183	Chr.7	15	291－324	0.7308	0.49
MCW0014	Chr.6	11	160－186	0.6774	0.51
MCW0067	Chr.10	6	178－186	0.6411	0.50
MCW0098	Chr.4	2	263－265	0.2915	0.29
LEI0166	Chr.3	7	354－376	0.6203	0.42
MCW0069	E46C08W18	9	158－176	0.7584	0.63
MCW0081	Chr.5	7	114－135	0.4702	0.27
ADL0112	Chr.10	4	124－132	0.4999	0.35
MCW0034	Chr.2	17	212－246	0.8301	0.67
MCW0111	Chr.1	12	96－120	0.7287	0.62
MCW0078	Chr.5	5	135－143	0.6650	0.50
MCW0206	Chr.2	11	221－247	0.7020	0.55
LEI0094	Chr.4	22	247－289	0.8849	0.78
MCW0248	Chr.1	5	215－223	0.6009	0.49
LEI0234	Chr.2	30	216－386	0.9205	0.81
MCW0330	Chr.17	7	258－290	0.7407	0.60
MCW0016	Chr.3	11	162－188	0.7198	0.56
MCW0104	Chr.13	19	190－232	0.8219	0.67
MCW0020	Chr.1	5	179－187	0.6420	0.50
MCW0165	Chr.23	3	114－118	0.5595	0.45
MCW0080	E18W15	15	266－282	0.7140	0.70
MCW0123	Chr.14	11	76－98	0.7907	0.67
Mean		9.86 (6.36)		0.6663 (0.1325)	0.52 (0.14)

Standard deviations for the mean number of alleles. He and PIC are given in parentheses.

performed with the RFLPscan software package (Scanalytics, Division of CSP, Billerica, USA).

1.4 Statistical analysis

(i) Genetic diversity. The total number of alleles, allele frequencies, average number of alleles per locus, observed (*Ho*) and expected heterozygosity (*He*) for each population across the loci were estimated with Microsatellite-Toolkit for Excel[10]. Polymorphism information content (*PIC*) for each locus and each breed was obtained according to Botstein et al.[11]:

$$PIC = 1 - \sum_{i=1}^{n} p_i^2 - 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} p_i^2 p_j^2$$

where n = the number of alleles; p_i= frequency of the allele i; p_j = frequency of the allele j.

(ii) Genetic differentiation. Population differentiation was estimated by fixation indices F_{IT}, F_{ST} and F_{IS} in the form of F, θ, and f, respectively, for each locus across populations according to the variance using FSTAT software (Version 2.9.3)[12]. The significance of the F-statistics was determined by permutation tests with the sequential Bonferroni procedure applied over loci. The extent of inbreeding was further studied with GENEPOP software[13] by estimating the F_{IS} values and their significance level within each of the populations.

(iii) Gene flow between populations (*Nm*). Pairwise F_{ST} values were computed for all combinations of the 16 populations using GENEPOP. Gene flow between populations, defined as the number of reproductively successful migrants per generation (*Nm*), was estimated based on the *n* island model of population structure. The estimate was based on the relationship $F_{ST}=1/(4Nm+1)$, where N is the effective population size, m is the migration rate, and F_{ST} is calculated as mean over loci.

(iv) Clustering of populations. An unrooted Neighbor-Joining cladogram[14] was obtained based on pair-wise kinship distance matrix between populations using the Neighbor-Joining program implemented in PHYLIP. A consensus tree, evaluated by 1000 bootstraps across the set of loci, was constructed.

2 Results

2.1 Genetic variability within populations

A total of 286 alleles were detected in 14 Chinese indigenous chicken breeds and two red jungle fowl subspecies with 29 microsatellite markers. Expected *He*, and mean *PIC* for each locus across 16 populations are listed in Table 2. The number of alleles per locus ranged from 2 (MCW0103 and MCW0098) to 30 (LEI0234) and the average number of the alleles observed in 29 microsatellite loci was 9.86±6.36. Across populations, locus MCW0098 had the lowest *He*, 0.2915 and the lowest PIC, 0.29, however the locus LEI0234 had the highest *He* and *PIC* value, 0.9205 and 0.81, respectively. The average number of alleles, expected and observed heterozygosity for each population across 29 loci are shown in Table 3. Across 29 loci, the lowest value 0.4532 of heterozygosity was obtained for the Gushi chicken breed, and the highest 0.6442 was found for Wannan Three-yellow. The overall expected heterozygosity of 14 Chinese indigenous chicken breeds and two red jungle fowl subspecies was 0.6708±0.0251.

As a measure of deviation from Hardy-Weinberg equilibrium, the F_{IS} value was calculated and typed-I error probability was computed. The levels of significance of deviations from Hardy-Weinberg equilibrium of each population for each locus (data not shown) were estimated and the number of populations deviated from HWE per locus ranged from 0 to 7 (Table 4).

2.2 Genetic differentiation among populations

Genetic differentiation was examined by fixation indices F_{IT}, F_{ST}, F_{IS} for each locus. Results of the F-statistics analysis for 29 microsatellite markers in 14 Chinese indigenous chicken breeds and two subspecies of red jungle fowl are presented in Table 4.

The fixation coefficients of subpopulations within the total population, measured as F_{ST} value, for the 29 loci varied from 0.111 (MCW0020) to 0.303 (MCW0081), with a mean of 0.167 (P<0.001), and all loci contributed significantly to this differentiation. The global deficit of heterozygotes across populations (F_{IT}) amounted to 15% (P<0.001). An overall significant deficit of heterozygotes (F_{IS}) of 1.5% (P<0.001) occurred in the analysed loci because of inbreeding within populations. Eight loci (MCW0216, MCW0295, MCW0222, MCW0014, LEI0094, MCW0248, LEI0234 and MCW0165) showed the significant deficit of heterozygotes and the F_{IS} value of these eight loci was 0.097 (P<0.01), 0.063 (P<0.08), 0.118 (P<0.001), 0.062 (P<0.01), 0.109 (P<0.001), 0.062 (P<0.05), 0.068 (P<0.001) and 0.128 (P<0.001), respectively. Twelve markers, to some extent, showed excess of heterozygotes, with a negative value.

Gene flow (*Nm*) and Reynolds' genetic distances (D_R) between each population pair are presented in Table 5. Reynolds' distance values varied between 0.036 (Xiaoshan chicken-Luyuan chicken pair) and 0.371 (Red

jungle fowl-Henan game chicken pair). The Nm value ranged from 0.583 (between Red jungle fowl and Gushi chicken pair) to 5.833 (between Xiaoshan chicken and Luyuan chicken pair).

Table 3 Mean number of alleles per locus and mean heterozygosity (*He* and *Ho*) for the 16 chicken populations

Population	Abbreviations	Number of alleles ($\bar{x}$ ± SD)	Average *He* ($\bar{x}$ ± SD)	Average *Ho* ($\bar{x}$ ± SD)
Xianju Chicken	XJ	3.90±2.19	0.5320±0.0324	0.5068±0.0152
Chahua Chicken	CH	4.34±2.02	0.5410±0.0357	0.4968±0.0154
Luyuan Chicken	LY	4.24±2.01	0.5784±0.0288	0.5320±0.0161
Gushi Chicken	GS	3.31±1.39	0.4532±0.0381	0.4489±0.0146
Tibetan Chicken	TC	5.24±2.63	0.6033±0.0337	0.6054±0.0149
Baier Chicken	BE	4.07±2.30	0.5172±0.0358	0.4847±0.0161
Dagu Chicken	DG	4.93±2.30	0.6271±0.0317	0.6404±0.0151
Henan Game	HG	3.72±1.77	0.5184±0.0375	0.5214±0.0162
Langshan Chicken	LS	4.10±1.93	0.5477±0.0303	0.6267±0.0142
Taihe Silkies	TS	4.48±1.99	0.5696±0.0305	0.5642±0.0146
Xiaoshan Chicken	XS	4.38±1.90	0.6039±0.0228	0.6125±0.0143
Beijing Fatty Chicken	BF	4.28±1.71	0.5405±0.0312	0.5626±0.0150
Huainan Partridge	HP	5.55±2.86	0.6194±0.0312	0.5727±0.0165
Wannan Three-yellow chicken	WTY	6.38±3.51	0.6442±0.0268	0.6053±0.0161
Gallus gallus gallus	GGG	4.79±1.84	0.6385±0.0311	0.5783±0.0180
Gallus gallus spadiceus	GGS	3.79±1.37	0.5379±0.0335	0.5356±0.0169

Table 4 The results from *F*-statistics analysis and the number of populations deviation from Hardy-Weinberg equilibrium

Locus	$F_{IT}=F$	$F_{ST}=\theta$	$F_{IS}=f$	Number of population deviation from Hardy-Weinberg equilibrium
MCW0103	0.143***	0.170***	–0.033	1
MCW0216	0.257***	0.177***	0.097**	3
MCW0295	0.190***	0.135**	0.063*	3
ADL0278	0.292***	0.277***	0.021	1
MCW0222	0.233***	0.131***	0.118***	5
MCW0037	0.250***	0.215***	0.044	2
ADL0268	0.150***	0.214***	–0.081	3
MCW0183	0.224***	0.221***	0.003	0
MCW0014	0.226***	0.174***	0.062**	3
MCW0067	0.077**	0.107***	–0.033	1
MCW0098	0.114**	0.123***	–0.010	0
LEI0166	0.236***	0.225***	0.014	0
MCW0069	0.143***	0.169***	–0.030	1
MCW0081	0.308***	0.303***	0.009	0
ADL0112	0.138***	0.217***	–0.101	3
MCW0034	0.108***	0.138***	–0.035	1
MCW0111	0.124***	0.132***	–0.010	2
MCW0078	0.136***	0.162***	–0.031	3
MCW0206	0.142***	0.116***	0.029	2
LEI0094	0.236***	0.143***	0.109***	7
MCW0248	0.190***	0.137***	0.062*	1
LEI0234	0.222***	0.166***	0.068***	4
MCW0330	0.207***	0.200***	0.022	2
MCW0016	0.185***	0.172***	0.016	3
MCW0104	0.117***	0.166***	–0.060	2
MCW0020	0.138***	0.111***	0.031	3
MCW0165	0.222***	0.107***	0.128***	3
MCW0080	0.140***	0.148***	–0.010	3
MCW0123	0.079***	0.117***	–0.043	3
Mean	0.179 (0.011) ***	0.167 (0.008) ***	0.015(0.011) **	

Mean estimates from jack-knife over loci, standard deviations are given in parentheses; *, $P<0.05$; **, $P<0.01$; ***, $P<0.001$.

Table 5 Reynolds' genetic distances, D_R (upper triangle) and the gene flow, *Nm* (lower triangle) between two red jungle fowl subspecies and 14 Chinese domestic chicken breeds

Population	XJ	CH	LY	GS	TC	BE	DG	HG	LS	TS	XS	BF	HP	WTY	GGG	GGS
XJ		0.244	0.140	0.155	0.072	0.103	0.080	0.290	0.137	0.108	0.191	0.222	0.081	0.122	0.227	0.207
CH	1.158		0.174	0.242	0.063	0.138	0.146	0.130	0.245	0.250	0.175	0.180	0.148	0.135	0.274	0.122
LY	1.435	0.918		0.114	0.080	0.108	0.123	0.184	0.170	0.148	0.036	0.059	0.084	0.056	0.246	0.181
GS	1.755	0.655	0.923		0.123	0.119	0.172	0.214	0.243	0.205	0.131	0.126	0.110	0.070	0.330	0.219
TC	4.740	4.576	1.640	1.283		0.053	0.042	0.150	0.134	0.117	0.110	0.126	0.114	0.063	0.203	0.106
BE	2.196	1.117	1.351	0.884	2.182		0.068	0.126	0.130	0.126	0.110	0.154	0.074	0.062	0.282	0.146
DG	1.918	1.198	1.755	1.145	2.407	1.627		0.173	0.101	0.124	0.176	0.196	0.079	0.093	0.231	0.191
HG	1.134	0.781	1.102	0.862	1.175	1.037	1.431		0.273	0.308	0.178	0.176	0.152	0.119	0.371	0.203
LS	1.448	0.835	1.271	0.805	1.467	1.374	1.543	1.062		0.065	0.232	0.250	0.076	0.163	0.247	0.216
TS	1.765	1.101	1.186	1.143	2.086	1.226	2.031	1.046	1.167		0.194	0.233	0.102	0.133	0.214	0.210
XS	1.425	0.928	5.833	0.968	1.586	1.527	2.080	1.134	1.298	1.313		0.054	0.138	0.076	0.230	0.148
BF	1.033	0.817	1.538	0.765	1.262	0.950	1.468	0.985	1.047	0.999	1.583		0.149	0.089	0.253	0.162
HP	2.462	1.315	1.968	1.170	3.110	1.951	2.955	1.435	2.058	1.792	1.863	1.571		0.050	0.265	0.178
WTY	2.503	1.405	1.953	1.482	3.212	2.298	3.555	1.512	1.729	2.124	2.348	1.598	4.750		0.241	0.170
GGG	0.734	0.720	0.679	0.533	1.046	0.601	1.019	0.583	0.725	0.822	0.730	0.651	0.932	1.229		0.167
GGS	1.081	1.293	1.109	0.695	1.638	1.079	1.653	0.913	0.954	1.332	1.289	0.936	1.807	2.078	1.040	

2.3 Clustering of populations

The Neighbour-Joining (NJ) tree derived from the kinship distances is given in Figure 1. The tree topology revealed two main clusters, although the relationships between breeds were not always supported by high bootstrap values. The heavy-body sized chicken breeds, Luyuan chicken, Xiaoshan chicken, Beijing Fatty chicken, Henan Game chicken, Huainan Partridge and Langshan chicken formed one branch, and it had a close genetic relationship between Xiaoshan chicken-Luyuan chicken pair and Chahua chicken-Tibetan chicken pair. These two pairs fell together with 99.3 percent boostraps and 88.8 percent boostraps, respectively. Chahua chicken and Tibetan chicken had closer genetic relationship with these two subspecies of red jungle fowl than other domestic chicken breeds. Two subspecies of red jungle fowl were on the outside of the phylogenetic tree. *G. gallus spadiceus* showed closer phylogenetic relationship with Chinese domestic chicken breeds than with *G. gallus gallus*.

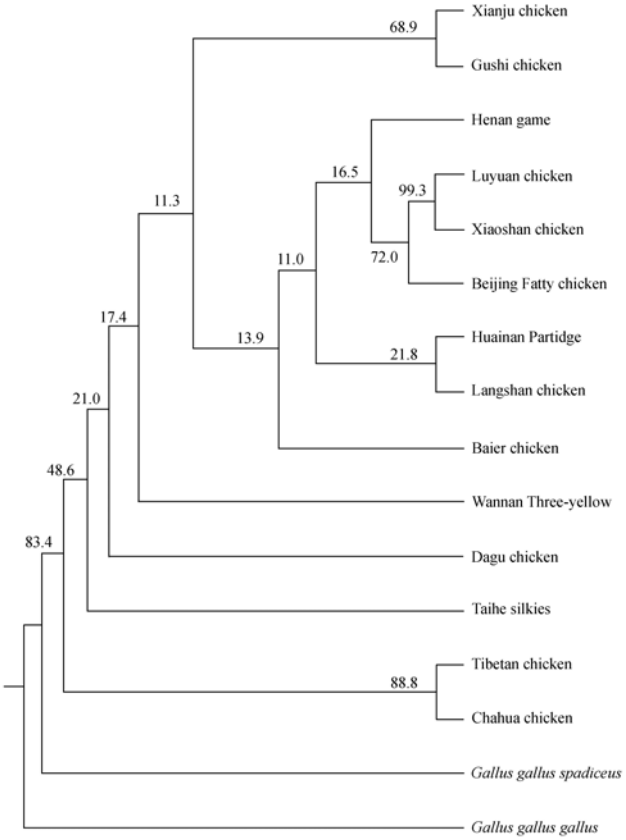


Figure 1 Dendogram of phylogenetic relationships among two subspecies of red jungle fowl and 14 Chinese domestic chicken breeds by using the neighbour-joining method. The numbers at the nodes are percentage bootstrap values from 1000 replications with resampled loci.

3 Discussion

3.1 Genetic variability within popualtions

29 microsatellite markers used in the present study were given by AVIANDIV EU-funded group, these markers were also used in the AVIANDIV EU-funded research project[8]. These markers are randomly distributed across 17 chromosomes or linkage groups in the chicken genome, so the data had certain commparability and representativeness. The *(PIC* value is a good measure of the

polymorphisms of gene fragment. When $PIC>0.5$, the locus is a highly polymorphic locus; when $0.25<PIC<0.5$, the locus is a medium polymorphic locus; when $PIC<0.25$, the locus is a low polymorphic locus[15]. Meanwhile, *PIC* value is related to the availability and utilization efficiency of a marker. The higher *PIC* value of the marker, the higher heterozygote frequency in one population, as well as the more genetic information it provides. In this study, 17 loci among 29 microsatellite loci exhibited high polymorphism, while 12 loci showed medium polymorphism. The mean *PIC* value across all loci exceeded 0.5, which could provide enough information for the assessment of genetic diversity.

Effective number of alleles is also a good measure of the genetic variation, especially in the conservation genetics study. Sometimes on its effect on populations is put more emphasis, but the effective number of alleles is easy to be affected by the sample size[16]. The average number of the alleles was 9.86 across 29 microsatellite loci in the present study, which indicated that the sample size was enough. On the other hand, this result also indicated that the polymorphism information provided by these 29 microsatellite loci in the 16 populations was rich; and the distribution of the allelic frequency was rather even. Therefore, using the effective number of alleles to analyze the genetic diversity is more effective and reliable.

Gene heterozygosity, also called gene diversity, is a suitable parameter for investigating genetic variation. Ott[17] gave a definition that a polymorphic locus must have at least 0.10 heterozygosity. All 29 microsatellite loci in this study had high polymorphism with a mean expected heterozygosity of 0.6663, showing a high degree of genetic diversity and relatively high selection potential. The mean expected heterozygosity can approximately reflect the variation of genetic structure. The Wannan Three-yellow chicken had the highest genetic variability, and the Gushi chicken had the lowest one. This might be due to the fact that the Wannan Three-yellow has just been founded in recent years with a large number of individuals and a broad distribution area. The genetic basis of the founder population of this breed is complicated. Some gene flow between Wannan Three-yellow and other breeds found in neighbouring regions possibly exists. This would explain the generally high *Nm* values of the Wannan Three-yellow and all other breeds. The special geographical conditions limit the Gushi breed to a relatively isolated region. The region is surrounded by mountains which may act as barriers to the gene flow. The breed therefore has less opportunity for genetic exchange with other populations as was indicated by the lower *Nm* values (from 0.533 to 1.755). *G. gallus gallus* had the higher genetic variability while *G. gallus spadiceus* showed the lower genetic variability, which was consistent with the location and present status of these two subspecies. *G. gallus gallus* is mainly distributed in the central and western mountainous areas of Thailand, its habitat distribution area is large and little selection has been performed on this breed. *G. gallus spadiceus* is the national second class conservation animal, distributed across Yunnan and Guangxi provinces in China, and usually habitats at evergreen broad-leaved forest, theropencedrymion, secondary bamboo forest and shrubs at altitude below 2500 m. *G. gallus spadiceus* is a relatively isolated population. In recent years, its distribution area as well as the population size decreased due to the destroyed forest, reclamation and hunting. The low genetic diversity detected in this study reflected this situation, which indicated that priorities for the conservation of the precious resource population should be conducted.

In the exact test for deviation from Hardy-Weinberg equilibrium, more or less populations showed significant deviation for all loci, except MCW0183, MCW0098, LEI0166 and MCW0081 (data not shown). Departures from HWE may be due to a variety of causes: small population size, assortative mating system (including inbreeding and outbreeding), selection, and existence of 'null alleles'.

3.2 Genetic differentiation among populations

In our study, on average, the genetic differentiation (F_{ST}) among breeds was 16.7% (Table 4), relatively high and extremely significant ($P<0.001$), which indicated that there is a great differentiation among the 16 chicken populations. It is clear that about 16.7% of the total genetic variation corresponds to differences of populations and the remaining 83.3% is the result of differences among individuals. All loci contribute to this differentiation significantly.

For the coefficient F_{IS}, which indicates the degree of departure from random mating, positive F_{IS} values mean a significant deficit of heterozygotes, while the negative F_{IS} values indicate an excess of heterozygous genotypes with respect to the expected value. In this study the high average of F_{IS} was 0.015. In addition, eight loci

(MCW0216, MCW0295, MCW0222, MCW0014, LEI0094, MCW0248, LEI0234 and MCW0165) showed significant deficit of heterozygotes. The values for most of the breeds, even if statistically significant, were not far from 0. Two reasons maybe contribute to the deficit of heterozygotes for these nine loci: first, the locus may be under selection (genetic hitchhiking effect) with some morphological or productive traits of selective interest; secondly, 'null alleles' may be present.

3.3 Phylogenetic relationships among red jungle fowls and Chinese domestic fowls

Environmental effects, historical process and life histories (e.g. mating system) may all, to some extent, constitute the genetic structure of populations. Geographical elements may owe to the close relationship for particular population pairs, for instance, Tibetan chicken and Chahua chicken, Xiaoshan chicken and Luyuan chicken. In the Neighbour-Joining tree, Tibetan chicken and Chahu clustered together with 88.8 percent boostraps, which indicated that they had a close genetic relationship. From geographical locations, Yunnan Province (Chahua chicken), is a neighbour to Tibetan, which made it convenient for these two chicken breeds to communicate with each other. Xiaoshan chicken and Luyuan chicken can be considered as genetically similar. The main origin locations Xiaoshan city and Zhangjiagang city, for Xiaoshan chicken and Luyuan chicken respectively, had the short geographical distance. Furthermore, the similar culture, living customs between these two places makes it easy to communicate with each other. The gene flow between these two breeds is the highest, being 5.833.

Bootstrapping values obtained in this analysis were between 11% and 99.3%, not always supporting the clustering result. However, according to the result from Figure 1, Chahua chicken and Tibetan chicken had a closer genetic relationship with these two subspecies of red jungle fowl, which are original native breeds between red jungle fowl and modern breeds that have had gene exchange with local red jungle fowls and have retained many primitive features. This result completely corresponed to the results reported by Cheng et al.[3] when analyzing the blood group, plasma protein polymorphic loci and DNA fingerprints of red jungle fowl and Chinese native fowl breeds. The red jungle fowl is the ancestor of the domestic fowl; it still maintains unique genetic characteristics due to its genetic stability and some other factors such as geographical distribution, ecological environment population isolation and so on. The domestic fowls have a long evolution history ever since they were domested seven to eight thousand years ago. They have experienced high intensity of artificial selection for different use under different natural and economical condition. Therefore, the genetic difference between domestic fowl and red jungle fowl becomes much larger[18]. Cheng et al.[3] and Wang et al.[19] concluded that Chahua chicken and Tibetan chicken have retained many primitive features of the red jungle fowl, as well as with the feathers of evolutional breeds. This also may be the reason why Tibetan chicken appeared as a mixture breed. However, it is more likely due to short of serve management during the domestication of these two breeds and little selection has been performed on these two breeds. The clustering result in this study further showed that Chahua chicken and Tibetan chicken are original native breeds between red jungle fowl and evolutional breeds. The evolutional dendrogram is as follows: evolutional breeds→primitive breeds (Chahua chicken and Tibetan)→red jungle fowl in China (*G. gallus spadiceus*)→red jungle fowl in Thailand (*G. gallus gallus*).

Some researchers have studied the origin of the Chinese domestic fowls and its relationship with red jungle fowl. Zheng[1] and Bo[2] suggested that the Chinese domestic fowls derived from red jungle fowl in China, which was consistent with the archaeological discovery that a great deal of domestic fowls fossils existed in Yangshao Culture Ruins of Yellow River Basin, and the Chinese domestic fowls might be reared seven to eight thousand years ago. Fumihito et al.[20,21] suggested that domestic fowls just derived from *G. gallus gallus* in Thailand and its surrounding region. Fu et al.[6] also held that the Chinese domestic fowls might derive from red jungle fowl in Thailand and its adjacent region. Cheng et al.[3] analyzed the blood group, plasma protein polymorphic loci and DNA fingerprints of red jungle fowl and Chinese native fowl breeds. The results supported the theory concerning the independent origin of Chinese native fowl breeds. Wang et al.[5] and Liu et al.[4] analyzed the polymorphisms of partial fragments of mitochondrial DNA of red jungle fowl and domestic fowl breeds. They drew the conclusion that domestic fowl breeds might derive from different *G. gallus* subspecies. In the current study, we did not get any samples from *G. gallus jabouillei* subspecies in China. But Liu et al.[22]

sequenced partial fragments of mitochondrial DNA cyt *b* and the control region of two Chinese *G. gallus* subspecies, showing that the genetic distance between the two *G. gallus* subspecies distributed in China was very close. The results from nuclear microsatellite data in this study concluded that red jungle fowl in Thailand and red jungle fowl in Thailand are different populations, the domestic fowls may originate from different subspecies of red jungle fowl, and Chinese domestic fowls have independent origin. The results in this study further supported the theory that domestic chickens have multiple origins. In the future, we should combine the evidence of molecular genetics and biochemical genetics with other aspects of information from morphology, archaeology and sociology, and then fully analyze genetic differentiation and phylogenetic relationships among red jungle fowls and domestic fowls. These results will be more useful for understanding the origin of Chinese domestic chickens.

In conclusion, all the 29 microsatellite loci examined in this study exhibited high or medium polymorphism, and the average number of the alleles was 9.86, which could provide enough effective and reliable information for the assessment of genetic diversity and phylogenetic relationships among red jungle fowls and Chinese domestic fowls. Across the 29 loci, the overall expected heterozygosity was 0.6663, with the lowest value obtained for the Gushi chicken, and the highest for the Wannan Three-yellow. *G. gallus spadiceus* in China showed the lower genetic variability and the priorities for the conservation of this precious population should be conducted. There is a great differentiation among the 16 chicken populations. On average, the genetic differentiation (F_{ST}) among the breeds was 16.7%, being relatively high and extremely significant (P<0.001), and all loci contribute to this differentiation significantly. An overall significant deficit of heterozygotes (F_{IS}) was relatively high, and eight loci showed significant deficit of heterozygotes. Chahua chicken and Tibetan chicken are original native breeds between red jungle fowl and evolutional breeds, with the evolutional dendrogram as follows: evolutional breeds→primitive breeds (Chahua chicken and Tibetan)→red jungle fowl in China (*G. gallus spadiceus*)→red jungle fowl in Thailand (*G. gallus gallus*). The results in this study indicated that red jungle fowl in Thailand and red jungle fowl in Thailand are different populations, the domestic fowls may originate from different subspecies of red jungle fowl, and Chinese domestic fowls have independent origin.

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